

Group 15 : Person Retrieval

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Abstract—This report details the implementation of a facial recognition system based on the Fisherfaces algorithm. The system leverages Principal Component Analysis (PCA) for dimensionality reduction and Linear Discriminant Analysis (LDA) to compute a discriminative feature space that maximizes class separability. The algorithm was applied to the ORL (AT&T) face database, which consists of 400 images from 40 different subjects. The implementation, developed in Python using scikit-learn and OpenCV, successfully projects faces into a low-dimensional "Fisherface" space. A query mechanism was developed to test the system, which calculates the Euclidean distance between a query image's feature vector and the database vectors to identify the closest matches, demonstrating high accuracy in subject identification.

Index Terms—Face Recognition, Fisherfaces, Principal Component Analysis (PCA), Linear Discriminant Analysis (LDA), ORL Dataset, Computer Vision, Biometrics

I. INTRODUCTION

Facial recognition is a prominent field within computer vision and biometrics, with applications ranging from security and surveillance to user authentication. A primary challenge is developing models that are robust to variations in lighting, facial expression, and pose.

Appearance-based models, which treat images as high-dimensional vectors, have proven effective. A well-known method, Eigenfaces, uses Principal Component Analysis (PCA) to find the components of maximum variance. However, PCA's effectiveness can be limited as the components of maximum variance may not be optimal for discriminating between different subjects.

The Fisherfaces method, proposed by Belhumeur, Hespanha, and Kriegman, improves upon this by applying Linear Discriminant Analysis (LDA) after an initial PCA step. LDA seeks to find a feature space that maximizes the ratio of between-class scatter (differences between subjects) to within-class scatter (variations of the same subject).

This project presents the design, implementation, and evaluation of a complete facial recognition system using the Fisherfaces algorithm. The system is trained on the ORL face database and provides an interactive interface to query and retrieve the most similar faces.

II. RELATED WORK

Facial recognition has been approached using various methods, broadly categorized into holistic, feature-based, and hybrid techniques.

A. Holistic Approaches (Eigenfaces)

The most famous holistic method is "Eigenfaces" [4], which relies solely on Principal Component Analysis (PCA). PCA identifies the directions of maximum variance in the training data, creating a "face space" spanned by eigenvectors (Eigenfaces). While computationally efficient, PCA's effectiveness is limited. The components of maximum variance are often dominated by external factors like lighting and background, not necessarily the features that distinguish one person from another. Our work uses PCA as an initial step, but not as the sole feature extractor.

B. Local Feature-Based Approaches (LBP)

In contrast to holistic methods, local feature-based techniques like Local Binary Patterns (LBP) analyze texture at a pixel level. LBP is highly robust to monotonic lighting changes and is effective at capturing fine-grained texture details [5]. As noted in our initial project exploration, LBP is a powerful descriptor. However, it can be sensitive to image rotation and may not capture the global structure of a face as effectively as holistic methods.

C. Fisherfaces (PCA + LDA)

The Fisherfaces method [3], which is the focus of this report, is a hybrid approach. It was developed to overcome the limitations of Eigenfaces. By applying Linear Discriminant Analysis (LDA) after PCA, it seeks to find a projection that not only reduces dimensionality but also maximizes class separability. It optimizes for a space where the differences between subjects are maximized while the differences within the same subject (due to expression, lighting) are minimized. This makes it theoretically superior to PCA-only methods for a classification task like face recognition, which is why we selected it for our implementation.

III. METHODOLOGY

The system was implemented in a Python environment, primarily using scikit-learn for machine learning and OpenCV for image processing. The overall methodology is divided into

four main stages: dataset preparation, preprocessing, model computation, and the recognition process. During dataset preparation, all images were organized and loaded in a consistent format. In preprocessing, the images were resized, converted to grayscale, and normalized to ensure uniform input. The model computation stage involved extracting Fisherfaces features and training the LDA model. Finally, during recognition, the system projected the query image into the same feature space and compared it with stored features to retrieve the most similar matches.

A. Dataset

The ORL (AT&T) face database was used. This standard dataset contains 400 grayscale images of 40 distinct subjects (10 images per subject). The images capture variations in expression, pose, and facial details (e.g., glasses).

B. Preprocessing

Before model training, all images were standardized to ensure consistency, a step noted in our project presentation.

- 1) **Grayscale Conversion:** Although the ORL dataset is already grayscale, this step is critical for color images to reduce complexity from three channels (RGB) to one (intensity).
- 2) **Resizing:** As per the provided code, each image was loaded and resized to a consistent 112x92 pixel resolution.
- 3) **Vectorization:** Each 2D image (112x92) was flattened into a 1D feature vector of 10,304 dimensions.
- 4) **Label Creation:** Each vector was associated with a numerical label (0 to 39) corresponding to the subject it represents.

C. The Fisherfaces Algorithm (PCA + LDA)

The core of the system is the two-stage Fisherfaces algorithm, which addresses the high dimensionality of the image data.

1) *Dimensionality Reduction with PCA:* The dataset has 400 samples (N) but 10,304 features (D), and only 40 classes (C). Applying LDA directly is problematic due to the "Small Sample Size" (SSS) problem, where the within-class scatter matrix (S_W) becomes singular (non-invertible).

To solve this, PCA is first applied to reduce the dimensionality. PCA finds a set of orthogonal axes that maximize the variance. As implemented in our code, the number of PCA components was dynamically set to $N - C$ (e.g., $400 - 40 = 360$), but capped at a maximum of 150. This step projects the data into a lower-dimensional space that is computationally feasible for LDA.

2) *Class-Separability with LDA:* After PCA, the N images are represented by their PCA-reduced vectors. Linear Discriminant Analysis (LDA) is then applied to this intermediate data. LDA's goal is to find the optimal projection vectors (Fisherfaces) that maximize the ratio of the between-class scatter matrix (S_B) to the within-class scatter matrix (S_W).

- **Within-Class Scatter (S_W):** Measures the variance of images belonging to the *same* subject (e.g., variations in expression). LDA aims to minimize this.
- **Between-Class Scatter (S_B):** Measures the variance between the means of different subjects. LDA aims to maximize this.

The resulting dimensionality of this final feature space is $C - 1$, or 39 dimensions. All 400 images in the database were transformed using this PCA+LDA model, and their final 39-dimensional weight vectors were stored.

D. Recognition Process

To identify a query face, the system (as seen in Cell 3) performs the following steps:

- 1) **Load Query Image:** An image is selected from the database via an `ipywidgets` dropdown.
- 2) **Transform Query:** The query image undergoes the *exact same* preprocessing and transformation pipeline as the training data: it is resized, flattened, projected using the `fitted_pca_model`, and then projected using the `fitted_lda_model`.
- 3) **Calculate Distance:** The Euclidean distance (`np.linalg.norm`) is computed between the query image's final 39-dimensional vector and every other vector in the stored database.
- 4) **Find Matches:** The distances are sorted, and the top 4 images with the smallest distances (excluding the query image itself) are identified as the best matches.

IV. RESULTS

The performance of our person-retrieval system was tested in a simple and practical way using the interactive query tool. In this tool, we select one image of a person (called the query image), and the system tries to find and display the top four most similar images from the entire database. This helps us check whether the system can correctly identify and retrieve images of the same individual under different conditions.

During testing, the system showed consistently high accuracy. For example, when we gave the system a query image of subject s7 (for instance, s7/5.pgm), the top four results returned by the system were also images of the same subject, such as s7/1.pgm, s7/6.pgm, s7/3.pgm, and s7/9.pgm. What makes this result even more impressive is that these matching images included different facial expressions and variations like wearing or not wearing glasses. This shows that the Fisherfaces-based model successfully focuses on the important and unique facial features of each person, while ignoring temporary or superficial changes.

Figure 1 provides an example of the system's output. It displays the query image on one side and the top four retrieved matches on the other side. In every test case we observed, the retrieved images belonged to the correct individual. The Euclidean distances for these matches were also very low, which means that the system placed the images of the same person very close to each other in the feature space. This



Fig. 1. Example query result from the implemented system. The leftmost image is the query. The following four images are the top matches retrieved from the database, which are all correctly identified as other images of same subject.

confirms that the model is not only identifying the correct person but also organizing the data in a meaningful way.

Overall, these results show that our person-retrieval system is reliable and effective. It can recognize people accurately even when their expressions change or when other minor variations are present. This demonstrates that the system is capable of handling real-world situations where a person may look slightly different in each image.

V. DISCUSSIONS

The results confirm the theoretical advantages of the Fisherfaces algorithm. While PCA (Eigenfaces) mainly focuses on capturing overall variance in the dataset, the addition of LDA transforms the feature space into one that is optimized specifically for distinguishing between different individuals. By reducing within-class scatter and increasing between-class scatter, the model becomes less affected by changes such as facial expressions, slight pose shifts, or the presence of glasses. At the same time, it becomes more responsive to the deeper, identity-related features that truly separate one person from another. These findings highlight why Fisherfaces consistently offers more stable and reliable performance in person-retrieval tasks compared to PCA alone.

A. Limitations and Challenges

Although our person-retrieval system performed well in controlled tests, several important limitations and challenges were observed during the project. These limitations highlight areas where the current approach struggles and where further improvements would be required for real-world deployment.

One major challenge we faced was the system’s sensitivity to lighting variations. Even though Linear Discriminant Analysis (LDA) is generally more stable than PCA, it is still affected when the lighting in the image is too strong, too weak, or uneven. In situations where half of the face is brighter than the other half, or when shadows fall across important facial regions, the model often fails to capture the true features of the person. This can cause the system to retrieve incorrect matches or rank the correct image lower than expected. During our observations, we noticed that many mismatches happened specifically when the query image had lighting conditions that

were very different from the images in the database. This clearly shows that lighting remains a major limitation and the system is not yet ready to handle real-world, uncontrolled environments.

Another significant limitation comes from the evaluation process itself. In our results, the system retrieves images from the same dataset on which it was fully trained. This means the model has already “seen” all the images during training, so naturally it performs well. However, this does not allow us to measure how well the system would work on completely new, unseen images of the same person. A more scientific and reliable approach would be to split the dataset into a separate training set and testing set—such as training the system on seven images per subject and testing it on the remaining three. With such a setup, we could compute meaningful accuracy metrics like Top-1 accuracy (whether the first retrieved match is correct) or Top-5 accuracy (whether the correct person appears anywhere in the top five results). Since our evaluation did not follow this structure, the results cannot be considered a complete measure of the system’s generalization ability.

Additionally, the ORL dataset itself brings certain limitations. It is considered a relatively “clean” dataset because the backgrounds are almost always uniform, the subjects are facing the camera, and the images are already well-aligned. There is little variation in pose, background clutter, or camera angle. Real-world person retrieval, however, is much more difficult. People may appear at different angles, partially occluded, in crowded environments, or with significant changes in lighting, hairstyle, or expression. If our current model were applied to such “in-the-wild” images, its accuracy would likely drop noticeably. To handle such challenging conditions, the system would require additional preprocessing steps, such as strong face-detection algorithms, reliable landmark detection, and geometric alignment (e.g., aligning eyes, nose, and mouth before feature extraction).

Overall, while our system shows promising results on the ORL dataset, these limitations clearly demonstrate the need for better handling of lighting, more rigorous evaluation through train-test splits, and stronger preprocessing techniques. These challenges guided us toward understanding the gap between

controlled experiments and real-world performance.

VI. CONCLUSION

This project successfully implemented a facial recognition system using the classical Fisherfaces (PCA+LDA) algorithm. The system, built with Python and `scikit-learn`, demonstrated its ability to project faces into a highly discriminative, low-dimensional feature space. The qualitative results from the interactive query tool show that the model can accurately identify subjects from the ORL database by finding the nearest neighbors in this feature space, validating the effectiveness of LDA for class separation.

Future work could focus on three main areas:

- 1) **Quantitative Evaluation:** Another important future improvement is the implementation of a proper quantitative evaluation using a formal train/test split. Instead of testing the system on the same images it was trained on, we plan to divide the dataset into two parts—one strictly for training and the other strictly for testing. This will allow us to calculate objective metrics such as Top-1 and Top-5 accuracy, precision, and retrieval consistency. Using this approach, we will be able to compare the Fisherfaces method more fairly against Eigenfaces and clearly see which model performs better in terms of generalization. By applying the same evaluation strategy to both models, we can understand their strengths, weaknesses, and suitability for real-world person retrieval tasks. This quantitative benchmarking will also help identify where further improvements, preprocessing steps, or feature-extraction enhancements are needed.
- 2) **Alternative Metrics:** Another future improvement is to replace the simple Euclidean distance with more advanced matching methods such as K-Nearest Neighbors (KNN). Instead of depending on just one closest image, KNN looks at the k nearest images and uses a voting system to decide the final match. This can make the retrieval process more stable and less sensitive to noise or small variations in lighting or expression. By analyzing multiple neighbors instead of one, the system can make more reliable decisions and reduce wrong matches, especially when the dataset becomes larger or more diverse.
- 3) **Hybrid Features:** Another promising direction is to explore a hybrid feature-extraction method that combines the holistic, discriminative power of Fisherfaces with the fine-grained texture information captured by Local Binary Patterns (LBP). By concatenating both feature vectors into a single representation, the system can learn richer details about each face. This hybrid descriptor has the potential to handle lighting changes, pose shifts, and small expression variations more effectively than using Fisherfaces alone. It would give the model both global structure and local texture cues, making person retrieval more stable and reliable in real-world, uncontrolled environments.

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